Math 271: Mathematical Methods (Fall 2026)
Tutorial 2
13 August, 2026

Solution Guidelines

Answer 1

Given,

$$\begin{array}{lll} & A = -A^\top & \\ \implies & A^\top = \left(-A^\top\right)^\top & \text{(Taking transpose on both sides)} \\ \iff & A^\top = -\left(A^\top\right)^\top = -A & \left(\because (kB)^\top = kB^\top \text{ for all } B \in \mathcal{M}^{m\times n}, \text{ for all } k \in \Re\right) \\ \therefore & A^\top = -A & \end{array}$$

Taking determinants both sides, we get

$$\begin{array}{lrl} \implies & \left|A^\top\right| = |-A| & \\ \implies & |A| = |-A| & \left(\because \left|B^\top\right| = |B| \text{ for all } B \in \mathcal{M}^{m\times n}\right) \\ \implies & |A| = (-1)^n|A| & \left(\because |kB| = (k)^n|B| \text{ for all } k \in \Re, \text{ for all } B \in \mathcal{M}^{m\times n}\right) \\ \implies & |A| = (-1)^3|A| = -|A| & \\ \implies & 2|A| = 0 & \\ \implies & |A| = 0 & \end{array}$$

Now, $|A| = 0$ implies by Theorem 3, A is not invertible. Furthermore, A is non-singular. We know that A is invertible $\Leftrightarrow$ A is non-singular. Thus, $\operatorname{rank}(A) < n$. Thus, the set of columns $\left\{A^1, A^2, A^3\right\}$ is a set of linearly dependent vectors. Hence $\exists x = (x_1, x_2, x_3) \in \Re^3$ s.t. $x \neq 0$ and $Ax = 0$ That is,

$$x_1 A^1 + x_2^a A^2 + x_3 A^3 = 0$$

Hence, the homogeneous system $Ax = 0$ admits non-trivial solutions.

Answer 2(a)

Write B through it columns $B = \left(B^1, B^2, \ldots, B^n\right)$.The j^{th} column of the product AB is given by AB^j, so $AB^j = 0 \quad (\forall j = 1, \cdots, n)$, i.e. every column of B solves the homogenous system $Ax = 0$ Define

$$N = \{x \in \mathbb{R}^n \mid Ax = 0\}$$

Then, for all $j = 1, \ldots, n$, we have $B^j \in N$. Hence, $\operatorname{rank}(B) \leq \operatorname{rank}(N)$. Thus,

$$\operatorname{rank}(A) + \operatorname{rank}(B) \leqslant \operatorname{rank}(A) + \operatorname{rank}(N)$$

Claim: $\mathrm{rank}(N) = n - \mathrm{rank}(A)$

Proof: Let $r = \mathrm{rank}(A)$.By Rank Theorem, $\mathrm{colrank}(A) = r$. Upon rearranging the columns & renumbering the unknows, we have $\mathcal{B}(A) := \{A^1, A^2, \ldots, A^r\}$ are a set of linearly independent columns. Then,

$$(\forall j \in \{r+1, r+2, ..., n\}): \quad Aj = t_{1j}A^1 + t_{2j}A^2 + \ldots + t_{rj}A^j$$

For all $j > r$ define a n dimensional column vector $z^j := (t_{1j}, t_{2j,\ldots} t_{rj}, 0, \ldots, 0, -1, 0, 0, \ldots, 0)^\top$, where -1 appears in the j^{th} position. So $Az^j = 0$.
Consider

$$\begin{aligned} z^{r+1} :=& \quad (t_{1j}, t_{2j}, \ldots, t_{rj} \quad -1, 0, 0 \ldots, 0, 0) \\ z^{r+2} :=& \quad (t_{1j}, t_{2j}, \ldots, t_{rj} \quad 0, -1, 0, \ldots, 0, 0) \\ \ldots& \\ z^n :=& \quad (t_{1j}, t_{2j}, \ldots, t_{rj} \quad 0, 0, 0, \ldots, 0, -1) \end{aligned}$$

Clearly, these $n-r$ vectors are linearly independent. Consider any z^j – there exists no $(n-r-1)$ scalars such that we can form a linear combination of $(n-r-1)$ remaining vectors to produce a 1 in the j^{th} position when all other vectors have a 0 in their j^{th} position.
Let $\tilde{x} \in \mathfrak{R}^n$ s.t. $A\tilde{x} = 0$, thus, $\tilde{x} \in N$. Define

$$y := \tilde{x} + \tilde{x}_{r+1}z^{r+1} + \ldots + \tilde{x}_n z^n$$

In the above $\tilde{x}_j$ is the j^{th} co-ordinate of $\tilde{x}$. Hence, we are forming a linear combination of any arbitrary $\tilde{x} \in N$ with $z^{r+1}, z^{r+2}, \ldots, z^n$ with weights $1, \tilde{x}_{r+1}, \tilde{x}_{r+1}, \ldots, \tilde{x}_n$. Then, $Ay = 0$
Furthermore, for $j > r \quad y_j = \tilde{x}_j + \tilde{x}_j z^j_j = \tilde{x}_j + \tilde{x}_j(-1) = 0$

So, $Ay = 0 \implies y_1A^1 + y_2A^2 + \ldots + y_rA^r = 0$. As $A^1, A^2, \ldots, A^r$ are linearly independent

$$\begin{aligned} \therefore& \quad y_1 = y_2 = \ldots = y_r = 0 \\ \therefore& \quad y = 0 \\ \therefore& \quad \tilde{x} = -\tilde{x}_{r+1}z^{r+1} - \tilde{x}_{r+2}z^{r+2} - \ldots - \tilde{x}_n z^n \end{aligned}$$

Hence for any arbitrary $\tilde{x} \in N$, we can express $\tilde{x}$ as a linear combination of linearly independent set of vectors, $\mathcal{B}(N) := \{z^{r+1}, z^{r+2}, \ldots, z^n\}$, hence they form a basis for N. Thus,

$$\mathrm{rank}(N) = |\mathcal{B}(N)| = n - r$$

Hence, $\mathrm{rank}(B) \leq \mathrm{rank}(N) = n - r$, thus,

$$\mathrm{rank}(A) + \mathrm{rank}(B) \leq \mathrm{rank}(A) + \mathrm{rank}(N) = r + (n - r) = n$$

Answer 2(b)

By (a), we have $\mathrm{rank}(A) + \mathrm{rank}(B) \leq n$. Furthermore, if $A \neq 0$, there exists $i \in \{1, \ldots, n\}$ such that $A^i \neq 0$, hence, $\mathrm{rank}(A) > 1$. Hence,

$$1 + \mathrm{rank}(B) < \mathrm{rank}(A) + \mathrm{rank}(B) \leq n$$

Hence,

$$\operatorname{rank}(B) \leq n-1 < n$$

Using the same logic as above,

$$\operatorname{rank}(A) \leq n-1 < n$$

Thus, both A and B are singular matrices. Furthermore, $\mathcal{B}(A) := \{A^1, \ldots, A^n\}$, $\mathcal{B}(B) := \{B^1, \ldots, B^n\}$ are linearly dependent set of vectors. Hence, there exists $x \in \mathfrak{R}^n$, $y \in R^n$ with $x \neq 0$ and $y \neq 0$, such that,

$$x_1 A^1 + x_2 A^2 + \ldots + x_n A^n = 0$$
$$y_1 B^1 + y_2 B^2 + \ldots + y_n B^n = 0$$

Hence, $Az = 0$, $Bz = 0$ have non-trivial solutions if $A \neq 0$ and $B \neq 0$ respectively.

Answer 2(a) Aliter

Some of you had suggested an aliter for part (a) along the lines of proving by contradiction. Here is a full written up version of the same:

Suppose it is not the case that $\operatorname{rank}(A) + \operatorname{rank}(B) \leq n$, then, $\operatorname{rank}(A) + \operatorname{rank}(B) \geq n+1$.

Common set-up

Again writing B through it columns $B = \left(B^1, B^2, \ldots, B^n\right)$.The j^{th} column of the product AB is given by AB^j, so $AB^j = 0 \quad (\forall j = 1, \cdots, n)$, i.e. every column of B solves the homogenous system $Ax = 0$ Define

$$N = \{x \in \mathbb{R}^n \mid Ax = 0\}$$

Then, for all $j = 1, \ldots, n$, we have $B^j \in N$.

Deriving Contradiction

Put $\operatorname{rank}(A) = r$ and $\operatorname{rank}(B) = s$ and suppose, contrary to the claim of the question, that $r+s \leq n$, we have $r+s > n \iff r+s \geq n+1$. Since the rank of B is the rank of its set of column vectors, we can pick s linearly independent columns of B, say $\mathcal{B}(B) := \{v^1, \ldots, v^s\}$; by the set-up, each $v^j \in N$, that is, $Av^j = 0$. Since $\operatorname{rank}(A) = r$, we can pick r linearly independent columns of A, say $\mathcal{B}(A) := \{A^{i_1}, A^{i_2}, \ldots, A^{i_r}\}$. Let $e^{i_1}, e^{i_2}, \ldots, e^{i_r}$ be the corresponding unit vectors in $\mathfrak{R}^n$, so that $Ae^{i_k} = A^{i_k}$.

Claim: The $r+s$ vectors $\{e^{i_1}, e^{i_2}, \ldots, e^{i_r}, v^1, v^2, \ldots, v^s\}$ are linearly independent.

Proof: Indeed, suppose

$$\alpha_1 e^{i_1} + \ldots, \alpha_r e^{i_r} + \beta_1 v^1 + \ldots + \beta_s = 0$$

Premultiply by A. Since $Av^j = 0$ for every $j \in \{1, \ldots, s\}$ this leaves,

$$\alpha_1 A^{i_1} + \alpha_2 A^{i_2} + \ldots + \alpha_r A^{i_r} = 0$$

and the independence of those columns of A forces $\alpha_k = 0$ for all $k \in \{1, \dots, r\}$. Thus,

$$\alpha_1 e^{i_1} + \dots, \alpha_r e^{i_r} + \beta_1 v^1 + \dots + \beta_s = \beta_1 v^1 + \dots + \beta_s = 0$$

and independence of the v^is forces $\beta_j = 0$ for all $j \in \{1, \dots, s\}$ This proves the claim.

So we have produced $r + s \geq n + 1$ linearly independent vectors in $\mathfrak{R}^n$. But any set of more than n vectors in $\mathfrak{R}^n$ is linearly dependent This is the desired contradiction. Hence, it is not the case that $r + s \geq n + 1$. Thus, $r + s \leq n$

Remarks: The construction proposed by some of you for the proof of contradiction involved proving the set $\{A^{i_1}, A^{i_2}, \dots, A^{i_r}, v^1, v^2, \dots, v^s\}$ as linearly independent set of vectors. This is false in general. Suppose

$$\beta_1 A^{j_1} + \cdots + \beta_r A^{j_r} + \alpha_1 v^1 + \cdots + \alpha_s v^s = 0.$$

Premultiplying by A gives

$$\beta_1 A A^{j_1} + \cdots + \beta_r A A^{j_r} = 0,$$

since $Av^i = 0$. To conclude $\beta_k = 0$, you would need $AA^{j_1}, \dots, AA^{j_r}$ to be linearly independent—meaning multiplying vectors formed by the columns of A by A again does not send non-zero combinations to zero, which is not guaranteed. (Using e^{j_k}, the same step produces $\beta_1 A^{j_1} + \cdots + \beta_r A^{j_r} = 0$, whose independence is guaranteed by the choice of basis columns.)

The claim is false in general. Take $n = 2$:

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad AB = 0.$$

Here $\mathrm{rank}(A) = \mathrm{rank}(B) = 1$.

- An independent column of A is $A^2 = (1, 0)^\top = e^1$.
- The only non-zero column of B is $v^1 = (1, 0)^\top = e^1$.

The proposed family is $\{e^1, e^1\}$, which is linearly dependent. This typically happens when $A^2 = 0$, where the vectors produced by A are themselves annihilated by A.

The standard proof avoids this: $Ae^2 = A^2 = e^1$ and $v^1 = e^1$, so $\{v^1, e^2\} = \{e^1, e^2\}$ is independent, yielding $r + s = 2 \leq n = 2$.

The structural reason. The set must consist of r vectors that A transforms into linearly independent vectors, together with s independent vectors satisfying $Av = 0$. This corresponds to splitting $\mathbb{R}^n$ into vectors that are annihilated by A and a complementary set of vectors that are not collapsed by A. Standard basis vectors e^{j_k} serve as suitable domain vectors since $Ae^j = A^j$ (any $u^1, \dots, u^r$ with $Au^1, \dots, Au^r$ independent would suffice). The columns A^{j_k} are output vectors, so they need not be linearly independent from vectors that are annihilated by A.

Answer 3

$(\Rightarrow)$: Suppose B has integer entries and

$$AB = BA = I$$

Since the determinant of an integer matrix $C, |C|$ is a sum of product of integers, $|A|$ and $|B|$ are integers, we have

$$\begin{aligned} &AB = I \\ \Rightarrow &|AB| = |I| = 1 \\ \Rightarrow &|A| \cdot |B| = 1 \quad (\because |CD| = |C| \cdot |D| \text{ for all } \quad C, D \in \mathcal{M}^{m\times n}) \end{aligned}$$

Two integers whose product is 1 are both 1 or both -1 , hence $|A| = 1$ or $|A| = -1$

$(\Leftarrow)$: Suppose $|A| = \pm 1$ In particular $|A| \neq 0$,
By Theorem 3, A is invertible, therefore,

$$A^{-1} = \frac{\text{adj}\, A}{|A|} = |A|\,\text{adj}\, A \quad (\because |A| = \pm 1)$$

Every entry of adj A is $\pm$ the determinant of an $(n-1)\times(n-1)$ sub-matrix of A, hence, an integer
So $B := |A|$ adj A is an integer matrix.
Thus, $AB = A \cdot (|A|\,\text{adj}\, A) = |A| \cdot A(\text{adj}\, A)$

$$= |A|^2 I = I$$

$$BA = |A|\,\text{adj}\, A \cdot A = |A|(\text{adj}\, A \cdot A) = |A|^2 I = I$$

Thus, B is an integer two sided inverse of A.

Remark: Writing $A^{-1} := |A| adj A$ is crucial because invertibility alone gives you an inverse, but the exercise asks for an inverse with integer entries, and mere invertibility says nothing about the entries.
In more detail, in the direction "$|A| = \pm 1 \implies$ there is an integer B with $AB = BA = I$":
From $|A| \neq 0$ you get, by Theorem 3, that A is invertible—but as a matrix over the reals. *A priori*, its inverse could have fractional entries (e.g., $A = \text{diag}(2, 1)$ is invertible, but $A^{-1} = \text{diag}(1/2, 1)$ is not an integer matrix). So "A is invertible" is not yet the statement we want.
What pins down the entries is the cofactor formula:

$$A^{-1} = \frac{\text{adj}\, A}{|A|}.$$

Every entry of $\text{adj}\, A$ is $\pm$ the determinant of an $(n-1)\times(n-1)$ submatrix of A, hence a sum of products of integers, and therefore an integer. Dividing by $|A| = \pm 1$ preserves that integrality—and since $1/|A| = |A|$ when $|A| = \pm 1$, the inverse is simply:

$$A^{-1} = |A| \cdot \text{adj}\, A,$$

visibly an integer matrix. Note that this is exactly where the hypothesis $|A| = \pm 1$ (as opposed to just $|A| \neq 0$) is used; with $|A| = 2$, the same formula would produce halves.
There is also a version of the argument that never mentions the inverse at all, which is arguably cleaner: set

$$B := |A| \cdot \operatorname{adj} A,$$

an integer matrix, and use the identity:

$$A \cdot \operatorname{adj} A = (\operatorname{adj} A) \cdot A = |A| I,$$

which holds for every square matrix with no division and no invertibility assumption. Then:

$$AB = |A|(A \operatorname{adj} A) = |A|^2 I = I$$

and likewise $BA = I$, since $|A|^2 = 1$. Thus, you obtain the existence of the integer two-sided inverse directly, and the invertibility of A emerges as a consequence rather than an input.

Answer 4

RTP:

$$(\operatorname{adj} A)^{-1} = \operatorname{adj}\left(A^{-1}\right) = \frac{A}{|A|}$$

Since A is non-singular, A is invertible, whence $|A| \neq 0$ and A^{-1} exists.

$$\operatorname{adj} A = |A| A^{-1}$$

We first prove $\operatorname{adj} A$ is invertible & $(\operatorname{adj} A)^{-1} = \frac{A}{|A|}$

$$\begin{aligned} &|A| I = A \cdot (\operatorname{adj} A) \\ \Rightarrow \quad &\frac{A}{|A|} \cdot (\operatorname{adj} A) = \frac{1}{|A|}(A \cdot \operatorname{adj} A) \\ &= \frac{1}{|A|}(|A| \cdot I) = I \end{aligned}$$

Furthermore

$$(\operatorname{adj} A) \cdot \frac{A}{|A|} = \frac{1}{|A|}((\operatorname{adj} A) \cdot A) = \frac{1}{|A|}(|A| I) = I$$

So $\frac{A}{|A|}$ is the two sided inverse of adj A.
since inverses are unique, (adj A) is invertible and $(\operatorname{adj} A)^{-1} = \frac{A}{|A|}$ For the other equality substitute A^{-1} instead of A in the cofactor-formula.

Answer 5

Refer Homework 2

Answer 6

Such an X always exists; if $n < m$ there are infinitely many.

Step 1: every system $Ax = c$ with $c \in \Re^n$ is solvable.
Let $c \in \Re^n$. The augmented matrix A_c is an $n \times (m+1)$ matrix, so its rank is at most the number of its rows, n (by the Rank Theorem, the rank is also the row rank, and there are only n rows). That is,

$$rank(A_c) \leq n$$

On the other hand A_c contains the columns of A and the vector c, so

$$rank(A_c) \geq rank(A) = n$$

Hence,

$$rank(A_c) = n = rank(A)$$

and Theorem 1 gives a solution $x^* \in \Re^m$ of the non-homogenous system $Ax = c$.

Step 2: construction of X.
Let $e^1, \ldots, e^n$ be the unit column vectors of $\Re^n$, i.e. the columns of I_n. By Step 1, choose for each i a vector $X^i \in \Re^m$ with

$$AX^i = e^i \qquad (i = 1, \ldots, n),$$

and let X be the $m \times n$ matrix with columns $X^1, \ldots, X^n$. Since the i-th column of the product AX is AX^i, the columns of AX are $e_1, \ldots, e_n$, that is,

$$AX = I_n.$$

Step 3: if $n < m$, there are infinitely many such X.
Because $rank(A) = n < m$, the m column vectors of A cannot be linearly independent (a linearly independent set of m vectors would give $rank(A) = m$), hence, there exists scalars $\{z_1\}_{i=1}^m$, not all zero, such that

$$z_1 A^1 + \ldots + z_m A^m = 0_m$$

That is, $z \in \Re^m$, $z \neq 0$, with $Az = 0$. Fix any non-zero row vector $u \in \Re^n$ and put

$$X(t) := X + t\,z\,u \qquad (t \in \Re),$$

where $z\,u$ is the $m \times n$ matrix with (j,i) entry $z_j u_i$. Then

$$AX(t) = AX + t\,(Az)\,u^\top = I_n + 0 = I_n$$

for every t, and the matrices $X(t)$ are pairwise distinct, because $zu^\top \neq 0$ (it has entry $z_j u_i \neq 0$ for suitable i, j) and $X(s) = X(t)$ would force $(s - t)\,zu^\top = 0$, i.e. $s = t$. Hence there are infinitely many right inverses.
